

Hybrid PAX–Photodiode Locking for ϕ_1 : Checkpoint Method and Results

Experimental checkpoint, 15 August 2026

Purpose. This checkpoint records the most repeatable current method for holding the ϕ_1 -controlled Poincaré-sphere azimuth. It is a *framework report*: the PAX monitor and photodiode (PD) may move in the full apparatus, so the measured local gains must be recalibrated, but the two-loop architecture and safety logic transfer directly.

1. Control architecture

The PAX supplies Stokes estimates (S_1, S_2, S_3) and the experimental azimuth is

$$u = \text{atan2}(S_3, S_2), \quad e_u = \text{wrap}(u_\star - u). \quad (1)$$

The PAX is deliberately not the fast actuator sensor. A PD at the final-output port supplies the fast error signal, while the PAX corrects the slow bias that makes a fixed PD level correspond to a changing polarization state. Figure 1 shows the separation of roles.

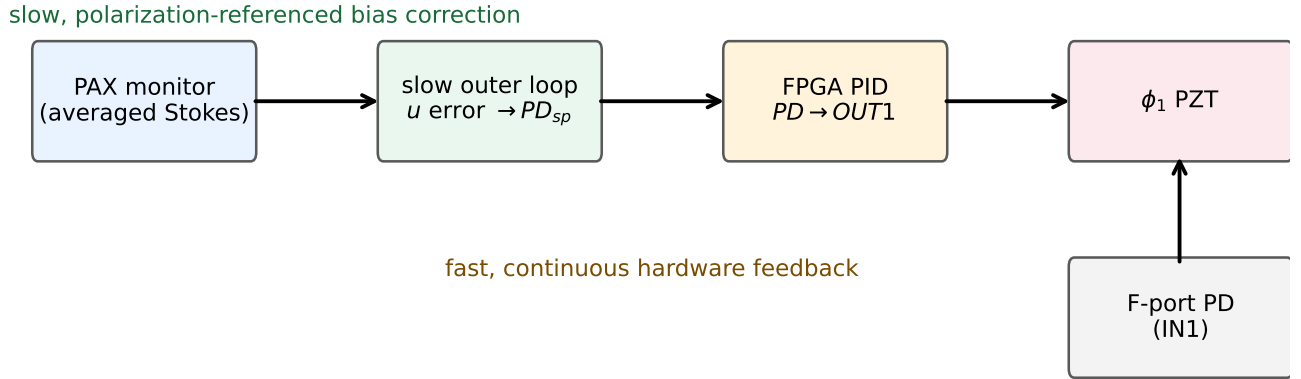


Figure 1: Hybrid control allocation. The FPGA PID drives $OUT1 \rightarrow \phi_1$ continuously; PAX changes only the FPGA PD setpoint at low bandwidth.

At each fresh local operating point, held sweeps measure

$$a = \frac{dPD}{dV_{OUT1}}, \quad b = \frac{du}{dV_{OUT1}}. \quad (2)$$

The signed FPGA proportional term is chosen for negative feedback, $P = -\text{sgn}(a)|P|$, and its magnitude is set by a normalized local loop gain $|Pa| = 0.22$. The integral unity-gain frequency is 5 Hz. The integrator is seeded at the present output, and a 1 s P-only check rejects a rail before integral action is enabled.

Every $T_o = 3$ s, an average of three PAX Stokes records updates the PD setpoint:

$$PD_{sp,k+1} = PD_{sp,k} + \text{clip}\left[g_o e_{u,k} \frac{a}{b}, -1.5 \text{ mV}, +1.5 \text{ mV}\right], \quad g_o = 0.12. \quad (3)$$

Since $du/dPD = b/a$, this converts a measured polarization error into the PD-setpoint shift that the FPGA must realize. Crucially, this is a *setpoint* trim: PAX never writes sparse voltage steps to the piezo.

2. Repeated short-hold evidence

Three independent 27 s holds using the settings above were selected because each completed under the autonomous experimental runtime limit and used fresh local calibration. The combined 30 PAX validation samples give a median $|\Delta u| = 0.0329$ rad, 90th percentile 0.131 rad, and median PD residual 1.59 mV. The distribution is centered ($\overline{\Delta u} = -0.0077$ rad), which is the key improvement over PD-only holds that could retain a substantial polarization bias.

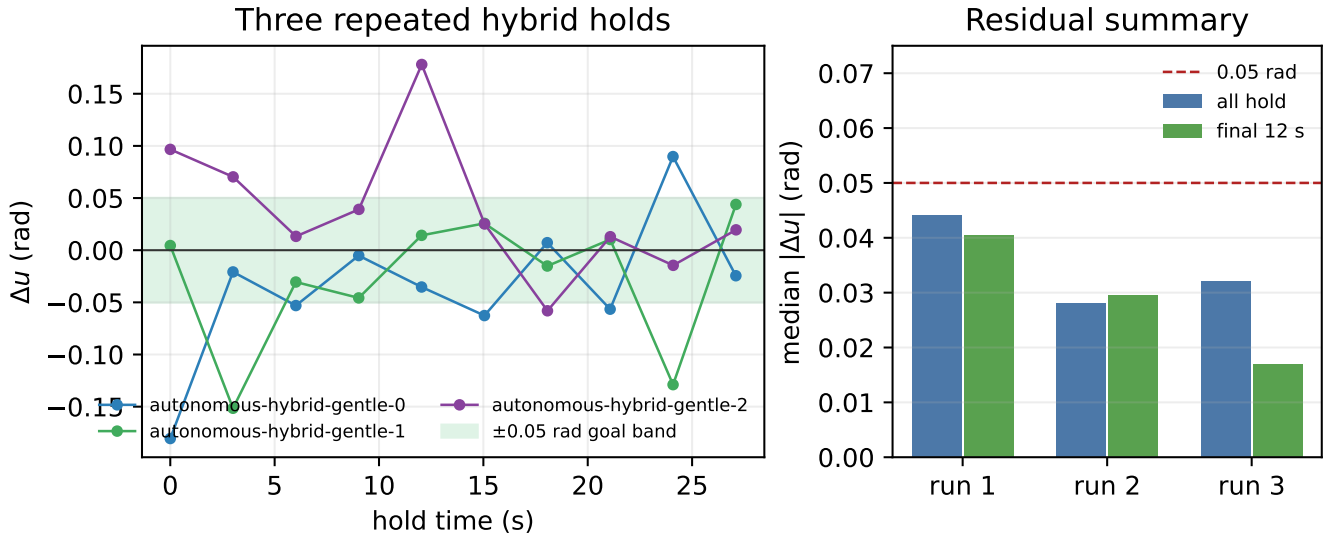


Figure 2: PAX validation of the retained hybrid method. The ± 0.05 rad band is a useful current target, not a claim of guaranteed long-duration performance.

Trial	median $ \Delta u $	final-window median	fraction $ \Delta u < 0.05$ rad
gentle-0	0.044 rad	0.057 rad	0.50
gentle-1	0.028 rad	0.026 rad	0.80
gentle-2	0.032 rad	0.020 rad	0.60

3. Transfer and retuning protocol

When either detector location changes, do not reuse the voltage-to-polarization or PD calibration numbers. Reuse the following sequence instead:

1. Place the system at the intended branch and run a fresh held ϕ_1 sweep. Confirm a locally monotonic PD flank and an appreciable $b = du/dV_{OUT1}$.
2. Measure a and b at that same branch. Set the sign of P from a ; begin with a conservative normalized gain below $|Pa| = 0.22$ and $I = 0$ during handoff.
3. Seed the FPGA integrator with the present output, perform the P-only rail check, then enable a low integral gain. Confirm that PAX u remains on the selected branch.
4. Enable the PAX outer trim with low gain and bounded PD-setpoint steps. Tune outer bandwidth only after the inner PD loop is stable; score changes by PAX $|\Delta u|$, not PD error alone.

Checkpoint conclusion. The experiment supports a practical division of labor: the PD/FPGA loop suppresses fast optical drift with smooth actuator output, while the slow PAX loop removes the otherwise persistent PD-to-polarization bias. The reported sub-0.05 rad short-hold results are promising and repeatable, but a separate 120–180 s confirmation remains required before claiming long-duration stability.